

# The Informant Questionnaire on Cognitive Decline in the Elderly (IQCODE): socio-demographic correlates, reliability, validity and some norms

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**SYNOPSIS** The IQCODE is a questionnaire which asks an informant about changes in an elderly person's everyday cognitive function. The questionnaire aims to assess cognitive decline independent of pre-morbid ability. In the present study, the IQCODE was administered to a sample of 613 informants from the general population. In addition, the questionnaire was administered to informants of 309 dementing subjects who had filled it out one year previously. A principal components analysis, using the general population sample, confirmed that the IQCODE measures a general factor of cognitive decline. The questionnaire was found to have high internal reliability in the general population sample ( $\alpha = 0.95$ ) and reasonably high test-retest reliability over one year in the dementing sample ( $r = 0.75$ ). The total IQCODE score, as well as each of the 26-items, was found to discriminate well between the general population and dementing samples. The correlation with education was quite small ( $r = -0.13$ ), indicating that contamination by pre-morbid ability is not a problem.

## INTRODUCTION

Dementia involves a decline in cognitive skills from a pre-morbid level. In assessing cognitive decline it is necessary to have estimates of both current and pre-morbid levels of ability. Assessment of current cognitive ability poses little problem, but records of pre-morbid ability are seldom available. Yet without a knowledge of pre-morbid ability there is a danger that individuals who have always been dull may be mistakenly diagnosed as demented, or conversely, that individuals who were pre-morbidly very bright may be classified as normal despite substantial cognitive decline.

In view of the general unavailability of records of pre-morbid ability, methods have been devised to estimate this indirectly. One such method involves the comparison of 'hold' *versus* 'don't hold' cognitive skills. The idea is that some cognitive skills tend to hold up well in

dementia whereas others tend to be more affected. The 'hold' skills are used to estimate pre-morbid ability. The 'hold' skill most commonly used for estimating pre-morbid ability is word reading (Nelson & O'Connell, 1978). While this approach is adequate for matching groups of individuals on pre-morbid ability, it can be quite inaccurate in individual cases, e.g. people suffering from a specific reading disability. A second approach is to use sociodemographic variables to estimate pre-morbid ability. A formula has been devised to predict ability from age, sex, education, race and urban/rural residence (Barona *et al.* 1984). Again, this approach works for group matching, but is insufficiently accurate for confident statements about decline in individual cases.

A possible solution to the problem of estimating cognitive decline is to use informants who have knowledge of both the subject's current and pre-morbid behaviour. Informant reports may also have greater ecological validity than the results of clinical tests, because they are based on observations of performance in everyday life rather than on artificial cognitive tasks.

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Informant reports are commonly used in clinical practice and have been incorporated in a number of formal assessment instruments such as the Blessed Dementia Scale (Blessed *et al.* 1968) and the CAMDEX (Roth *et al.* 1986). However, these instruments have used informants primarily to assess current skills rather than decline from a pre-morbid level.

The present paper reports on an instrument designed to measure cognitive decline using an informant's report. This instrument began as a 26-item interview in which an informant was asked to rate the degree of change over a ten-year period in various aspects of an elderly person's memory and intelligence (Jorm & Korten, 1987). A ten-year period was chosen because it provides a convenient mental landmark for rating and because the duration from dementia onset to death is generally less than a decade (Barclay *et al.* 1985; Diesfeldt *et al.* 1986). The total decline score derived from this interview was found to correlate highly with the Mini-Mental State Examination (Folstein *et al.* 1975), yet to have low correlations with indicators of pre-morbid ability. Later, Jorm *et al.* (1989) developed a self-administered questionnaire version of the interview which might serve as a useful screening instrument for dementia or as a component of a full diagnostic assessment. Informants were asked to rate performance on various everyday cognitive tasks as 'much better', 'a bit better', 'not much change', 'a bit worse', or 'much worse'. This questionnaire was called the Informant Questionnaire on Cognitive Decline in the Elderly and given the acronym IQCODE (where *IQ* stands for *Informant Questionnaire*). The IQCODE items are reproduced in Appendix 1. Jorm *et al.* (1989) administered the questionnaire by mail to a sample of 362 informants who were members of the Alzheimer's Disease and Related Disorders Society (ADARDS). A psychometric analysis of the data showed that the IQCODE measured a broad general factor of cognitive decline with high internal reliability. Furthermore, it had virtually zero correlation with the dementing person's education and occupational status. A small validation study using elderly people in residential care showed a high correlation with the Mini-Mental State, as found previously for the interview version.

Reported here are further data on the re-

liability and validity of the IQCODE, collected from a general population sample of informants and from a one-year follow-up of the earlier ADARDS sample.

## METHOD

### General population sample

A list of 4000 names was taken from the electoral roll for the Australian Capital Territory. Registration on the electoral roll is compulsory for all Australians aged 18 and over. These individuals were sent a letter asking whether they had had regular contact throughout the last 10 years with a person now aged 70 or over and, if so, whether they were willing to fill out a questionnaire about them. Replies were received from 1658 individuals and 138 letters were returned to sender. Of the 1658 replies, 815 said they knew a suitable elderly person and were willing to fill out a questionnaire, 37 knew of a suitable person but were not willing to do a questionnaire, and 761 did not know a suitable elderly person. Because some of the 815 willing and able to do the questionnaire responded tardily, questionnaires were sent to only 786 of them, and the return rate of these questionnaires was 78% ( $N = 613$ ). These 613 informants formed the basis for the analyses reported below. In planning the survey, we had estimated a final sample of 400, so this response was somewhat better than anticipated. These informants, and the elderly subjects they reported on, could not be claimed to be representative of the general Australian population. Rather, they were a volunteer sample drawn from the general population. Table 1 shows the age breakdown for the elderly reported on in this sample, compared to

Table 1. *Percentage distribution by age and sex for the general population sample compared to the Australian elderly population*

| Age group | General population sample |             | Australian elderly population |             |
|-----------|---------------------------|-------------|-------------------------------|-------------|
|           | Males (%)                 | Females (%) | Males (%)                     | Females (%) |
| 70-74     | 12.6                      | 19.6        | 18.8                          | 23.8        |
| 75-79     | 11.7                      | 20.6        | 11.9                          | 17.2        |
| 80-84     | 5.5                       | 11.7        | 6.0                           | 10.7        |
| 85+       | 4.7                       | 13.5        | 3.1                           | 8.5         |
| Total     | 34.6                      | 65.4        | 39.8                          | 60.2        |

the breakdown for the Australian elderly population. It can be seen that the 85+ age-group is over-represented in the present sample and the 70–74 year-old group is under-represented. Also, females are somewhat over-represented compared to males. Informants were asked if the subject they were reporting on had been diagnosed as suffering from dementia or Alzheimer's disease. This had occurred in 5.7% of the sample.

### Follow-up of ADARDS sample

This sample has been described previously by Jorm *et al.* (1989). Basically, it consisted of 362 informants who are members of ADARDS and could report on a dementing person. Of this sample, 27% were under 70 years, 18% were 70–74, 25% were 75–79, 20% were 80–84, and 9% were 85+. One year after the IQCODE was sent out, the informants were sent a follow-up questionnaire asking whether the dementing person was still alive. If the dementing person was alive, the informant was asked about that person's current living arrangements and to again fill out the IQCODE. Replies were received from 309 informants, with 49 of these reporting that the dementing person had died.

## RESULTS

### Scoring

Several ways of scoring the IQCODE are possible, including summing the ratings over all items to give a maximum score of 130 or dichotomizing the items before summing to give a score out of 26. The method used here is to average the ratings over the 26-items. Unlike the other methods, this one has the advantage of yielding an immediately interpretable score because 1 represents considerable improvement in cognitive function, 3 represents no change, and 5 represents considerable deterioration. Scores are averaged to two decimal places to give a finer gradation of cognitive change.

### Socio-demographic correlates

The IQCODE score was correlated with various sociodemographic variables in the general population sample. The correlation with age was 0.33, implying that older people have greater decline scores. There was a correlation with sex,  $r = 0.10$ , such that females had greater decline

scores than males. Although statistically significant with the large sample,  $P = 0.008$ , the size of the relationship is small.

To see whether the pre-morbid ability of the elderly subject might affect the IQCODE as completed by the informant, the correlation with education was calculated. The correlation was  $-0.06$  with age of leaving school and  $-0.13$  with highest qualification obtained (measured on an eight-point scale ranging from 'no schooling' to 'completed tertiary college or university'). Again, the latter correlation is significant,  $P = 0.001$ , because of the large sample size. This correlation means that the poorly educated tended to have greater decline scores.

Correlations were calculated between the IQCODE and various characteristics of the informants. There were near-zero relationships with the informant's age, sex and the length of time since the informant lived with the subject.

### Reliability

Cronbach's alpha was calculated for the general population sample to assess internal-consistency reliability. A value of 0.95 was found, similar to the alpha of 0.93 previously reported with the ADARDS sample.

To assess stability of the scale, the one-year test-retest correlation was calculated for the ADARDS sample. This correlation was 0.75. As would be expected with dementing individuals, the mean score on the IQCODE over this period increased from 4.67 (S.D. = 0.37) to 4.76 (S.D. = 0.34), and this change was statistically significant,  $t(252) = 5.97$ ,  $P < 0.001$ .

### Validity

The IQCODE is supposed to provide a unidimensional measure of cognitive decline. To assess the factorial composition of the questionnaire, a principal components analysis was carried out on the 26-items using the general population sample. Table 2 shows the item loadings of the first four principal components. The first principal component was found to account for much more of the variance than any of the others. All items loaded on this first principal component, but the loadings for items 15 and 16 were lower than for the other items. The second principal component had high

Table 2. *Principal-component analysis of IQCODE: item loadings of first four components*

| Item          | Component |       |       |       |
|---------------|-----------|-------|-------|-------|
|               | 1         | 2     | 3     | 4     |
| 1             | 0.69      | -0.03 | -0.21 | -0.06 |
| 2             | 0.72      | -0.14 | -0.39 | -0.08 |
| 3             | 0.74      | -0.15 | -0.33 | -0.00 |
| 4             | 0.78      | -0.21 | -0.18 | 0.01  |
| 5             | 0.78      | -0.19 | -0.17 | 0.03  |
| 6             | 0.66      | -0.04 | -0.11 | -0.15 |
| 7             | 0.73      | 0.10  | -0.25 | -0.09 |
| 8             | 0.78      | -0.06 | -0.33 | -0.08 |
| 9             | 0.72      | -0.16 | -0.23 | 0.12  |
| 10            | 0.73      | -0.24 | -0.18 | 0.13  |
| 11            | 0.62      | -0.20 | 0.34  | 0.16  |
| 12            | 0.74      | 0.01  | 0.23  | 0.06  |
| 13            | 0.67      | -0.17 | 0.42  | 0.12  |
| 14            | 0.71      | -0.09 | 0.36  | -0.03 |
| 15            | 0.28      | 0.78  | -0.16 | 0.39  |
| 16            | 0.40      | 0.76  | -0.05 | 0.28  |
| 17            | 0.62      | 0.33  | 0.22  | -0.30 |
| 18            | 0.66      | 0.32  | 0.16  | -0.47 |
| 19            | 0.68      | 0.25  | 0.15  | -0.42 |
| 20            | 0.75      | 0.02  | 0.05  | -0.20 |
| 21            | 0.59      | 0.44  | 0.07  | 0.02  |
| 22            | 0.73      | -0.13 | 0.23  | 0.26  |
| 23            | 0.74      | -0.07 | 0.11  | 0.26  |
| 24            | 0.74      | -0.08 | 0.20  | 0.23  |
| 25            | 0.81      | -0.04 | 0.04  | 0.09  |
| 26            | 0.77      | -0.02 | 0.14  | -0.01 |
| Eigenvalue    | 12.54     | 1.98  | 1.37  | 1.07  |
| % of variance | 48.2      | 7.6   | 5.3   | 4.1   |

loadings only for items 15 and 16, which asked about memory for the distant past. Components 3 and 4 had no high loadings and could not be meaningfully interpreted. The principal components analysis therefore indicated that the IQCODE is largely measuring a broad general factor of cognitive decline, with items 15 and 16 being poorer measures of this factor than the rest of the questionnaire.

If the IQCODE is valid, it should discriminate clearly between dementing and normal groups. Since the ADARDS sample involved reports on dementing subjects, and the general population sample involved reports on subjects who were predominantly non-dementing, the extent to which the scale discriminated these two groups was examined. The mean score of the ADARDS sample (at time 1) was 4.67 (s.d. = 0.39) compared to 3.37 (s.d. = 0.51) for the general population sample. This difference was statistically significant,  $t(967) = 41.69$ ,  $P < 0.001$ . Fig. 1 shows the frequency distributions of the

IQCODE scores for the two groups. It can be seen that the two groups are well separated by the questionnaire. When the discrimination of each IQCODE item was examined, the two groups were found to differ on every item ( $P < 0.001$ ). These criterion groups are not ideal for determining the sensitivity and specificity of the IQCODE because the general population sample would have contained some demented individuals and the ADARDS sample may have contained some normal ones. Nevertheless, if a cutting point of 4+ was used to define likely dementia, the questionnaire would classify 92.7% of the ADARDS sample as demented and 88.1% of the general population sample as normal.

The predictive validity of the IQCODE was assessed in the ADARDS sample by seeing whether it could discriminate those subjects who died or became institutionalized at follow-up. Such an association was expected if severe dementia is predictive of death and institutionalization. However, given that there are many other potential causes involved in these outcomes, they provide only a weak test of the questionnaire's validity. It was found that the IQCODE score at time 1 did not differ between those who subsequently died ( $\bar{X} = 4.74$ , s.d. = 0.44) and those who survived ( $\bar{X} = 4.67$ , s.d. = 0.37). However, the IQCODE scores of subjects who subsequently became institutionalized ( $\bar{X} = 4.72$ , s.d. = 0.19) were worse than for subjects who remained in the community ( $\bar{X} = 4.44$ , s.d. = 0.41),  $t(128) = 5.32$ ,  $P < 0.001$ .

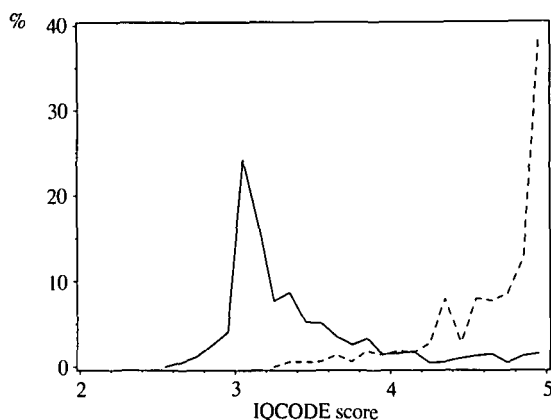


Fig. 1. Frequency distributions of IQCODE scores for the general population and ADARDS samples. (—, General population; ----, ADARDS.)

### Education bias

An intended advantage of the IQCODE over conventional screening instruments is that it focuses specifically on cognitive decline and is uncontaminated by pre-morbid ability. However, in the general population sample the IQCODE had a small, but statistically significant, correlation with education, so it is interesting to consider whether this was due to test bias or to a real tendency for the poorly educated to experience greater cognitive decline in old age.

Jensen (1980) has defined test bias in terms of 'systematic errors in the *predictive validity* or the *construct validity* of test scores of individuals that are associated with the individual's group membership' (p. 374). Thus, the IQCODE could be considered biased if its validity differed for more- and less-educated groups. Accordingly, the sample was divided into a group with primary education or less ( $N = 163$ ) and a group with secondary education or more ( $N = 443$ ). consistent with the correlation reported earlier, the less-educated group tended to score worse than the better-educated group (means of 3.50 and 3.33 respectively) and this difference was statistically significant,  $F(1, 602) = 6.99$ ,  $P = 0.008$ .

To assess construct validity, a principal components analysis was carried out separately for each of these groups. This revealed a similar structure, with the first principal component accounting for 45% of the variance in the more-educated group and 54% in the less-educated group. The loadings on this component were compared by a coefficient of congruence (Reynolds, 1982) to see if they were identical. The coefficient had a value of 0.998, indicating that the same principal component was being measured in both groups.

The correlation with age may also be seen as a validity coefficient. This was identical ( $r = 0.32$ ) for the two groups.

Because IQCODE scores should be worse for people in institutional care, the difference between community and institutional residents was assessed. The institutional residents tended to perform worse, but importantly, there was no interaction with education, showing that the IQCODE discriminated equally well for the more- and less-educated groups. A similar lack of interaction with education was found when

people medically diagnosed as demented (according to the informant's report) were compared to those not so diagnosed.

Coefficient alpha was also calculated and found to be high for both groups (0.96 for the less-educated v. 0.95 for the more educated).

In summary, there was no evidence for differential validity or reliability between the more- and less-educated groups, leading to the conclusion that the IQCODE is not biased against the poorly educated.

### Norms

While the general population sample may not be fully representative, it is sufficiently comprehensive to provide some preliminary norms. Appendix 2 gives percentile rank equivalents of IQCODE scores for various age groups. These percentiles show the percentage of subjects who have an equal or lesser cognitive decline score. In other words, a high percentile rank indicates greater decline. It is interesting to note that the differences between the age groups results because of a general lowering of scores at all points in the distribution rather than because a pathological group with extreme scores becomes more numerous.

### DISCUSSION

The present results confirmed in a general population sample that the IQCODE measures a general factor of cognitive decline, thus supporting its construct validity. There was also evidence of concurrent validity in that the questionnaire discriminated well between the ADARDS and general population samples. However, it must be emphasized that these were not ideal criterion groups, since some of the general population sample would have been demented and some of the ADARDS sample may have been non-demented. Adequate determination of the sensitivity and specificity of the IQCODE for screening purposes must await the use of accurately diagnosed criterion groups. Some evidence of predictive validity was also found, based on the assumption that severe dementia increases the probability of death or institutionalization. Dementing subjects who subsequently moved into institutional care tended to have worse scores than those who remained in the community. However, the

IQCODE did not discriminate dementing subjects who subsequently died from those who survived.

The present study also confirmed in a general population sample that the IQCODE has high internal reliability. In addition, the re-test reliability of the questionnaire was assessed over one year in the dementing sample and was found to be reasonably high. If anything, the test-retest reliability is likely to be lower in a dementing sample than in the general population, because many dementing subjects score at or near the IQCODE's ceiling and many also change their score because of the dementia process.

The unifactorial structure of the IQCODE and its high internal reliability indicate that it could possibly be shortened without too much loss of information. However given that research with the IQCODE is only in its early stages, the development of a short form might be premature. It would encourage use of a subset of the 26 items and thereby prevent collection of data on which to base the selection of an optimal set of items.

The IQCODE was found to correlate with age in the general population sample. In many ways this is not surprising, because traditional cognitive tests also correlate with age. However, since the IQCODE is a measure of decline over ten years rather than current cognitive status, a correlation with age implies that the average rate of decline increases with age. If cognitive function declined as a linear function of age, then there would be a negative correlation with traditional tests, but a zero correlation with the IQCODE. This is because the amount of change over a 10-year period would be equal for both the young old and the old old. By contrast, the present results are consistent with a downwardly accelerating curve relating age to cognitive function. It must be emphasized, of course, that this age relationship is from cross-sectional data and may reflect either ageing or cohort effects and is based on grouped results which may not reflect the shape of individual trajectories.

A major reason for developing the IQCODE was to overcome the contamination with pre-morbid ability which plagues conventional cognitive screening tests for the elderly. Despite this aim, the questionnaire was found to have a small, but statistically significant, correlation with education. Such a correlation might result

because of bias in the test or because the poorly educated really tend to decline more in old age. To check for test bias, the general population sample was divided into more-educated and less-educated groups and the IQCODE's validity was examined separately in each. A biased test would be expected to have differential validity as a function of education. However, no evidence of test bias was found. Therefore, consistent with Bank & Jarvik's (1978) longitudinal observations, it seems possible that the poorly educated tend to experience greater cognitive decline in old age.

Thanks are due to ADARDS (NSW) for help with the research. Helpful comments on an earlier draft of the manuscript were provided by Helen Christensen, Scott Henderson and Ailsa Korten. Penny Evans and Susan Slater typed the manuscript and John Tucker produced the Figure.

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## APPENDIX 1

## Informant Questionnaire on Cognitive Decline in the Elderly (IQCODE)

(Note: As administered in the studies reported here, the IQCODE was preceded by questions on sociodemographic characteristics and physical health.)

Now we want you to remember what your friend or relative was like 10 years ago and to compare it with what he/she is like now. 10 years ago was in 19—. Below are situations where this person has to use his/her memory or intelligence and we want you to indicate whether this has improved, stayed the same, or got worse in that situation over the past 10 years. Note the importance of comparing his/her present performance *with 10 years ago*. So if 10 years ago this person always forgot where he/she had left things, and he/she still does, then this would be considered 'Hasn't changed much'. Please indicate the changes you have observed by *circling the appropriate answer*.

Compared with 10 years ago how is this person at:

|                                                                                          | 1           | 2            | 3               | 4           | 5          |
|------------------------------------------------------------------------------------------|-------------|--------------|-----------------|-------------|------------|
| 1. Recognizing the faces of family and friends                                           | Much better | A bit better | Not much change | A bit worse | Much worse |
| 2. Remembering the names of family and friends                                           | Much better | A bit better | Not much change | A bit worse | Much worse |
| 3. Remembering things about family and friends e.g. occupations, birthdays, addresses    | Much better | A bit better | Not much change | A bit worse | Much worse |
| 4. Remembering things that have happened recently                                        | Much better | A bit better | Not much change | A bit worse | Much worse |
| 5. Recalling conversations a few days later                                              | Much better | A bit better | Not much change | A bit worse | Much worse |
| 6. Forgetting what he/she wanted to say in the middle of a conversation                  | Much better | A bit better | Not much change | A bit worse | Much worse |
| 7. Remembering his/her address and telephone number                                      | Much better | A bit better | Not much change | A bit worse | Much worse |
| 8. Remembering what day and month it is                                                  | Much better | A bit better | Not much change | A bit worse | Much worse |
| 9. Remembering where things are usually kept                                             | Much better | A bit better | Not much change | A bit worse | Much worse |
| 10. Remembering where to find things which have been put in a different place from usual | Much better | A bit better | Not much change | A bit worse | Much worse |
| 11. Adjusting to any change in his/her day-to-day routine                                | Much better | A bit better | Not much change | A bit worse | Much worse |
| 12. Knowing how to work familiar machines around the house                               | Much better | A bit better | Not much change | A bit worse | Much worse |
| 13. Learning to use a new gadget or machine around the house                             | Much better | A bit better | Not much change | A bit worse | Much worse |
| 14. Learning new things in general                                                       | Much better | A bit better | Not much change | A bit worse | Much worse |
| 15. Remembering things that happened to him/her when he/she was young                    | Much better | A bit better | Not much change | A bit worse | Much worse |
| 16. Remembering things he/she learned when he/she was young                              | Much better | A bit better | Not much change | A bit worse | Much worse |
| 17. Understanding the meaning of unusual words                                           | Much better | A bit better | Not much change | A bit worse | Much worse |
| 18. Understanding magazine or newspaper articles                                         | Much better | A bit better | Not much change | A bit worse | Much worse |

Appendix 1 (cont.)

|                                                                                                                                            | 1           | 2            | 3               | 4           | 5          |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------|-----------------|-------------|------------|
| 19. Following a story in a book or on TV                                                                                                   | Much better | A bit better | Not much change | A bit worse | Much worse |
| 20. Composing a letter to friends or for business purposes                                                                                 | Much better | A bit better | Not much change | A bit worse | Much worse |
| 21. Knowing about important historical events of the past                                                                                  | Much better | A bit better | Not much change | A bit worse | Much worse |
| 22. Making decisions on everyday matters                                                                                                   | Much better | A bit better | Not much change | A bit worse | Much worse |
| 23. Handling money for shopping                                                                                                            | Much better | A bit better | Not much change | A bit worse | Much worse |
| 24. Handling financial matters, e.g. the pension, dealing with the bank                                                                    | Much better | A bit better | Not much change | A bit worse | Much worse |
| 25. Handling other everyday arithmetic problems, e.g. knowing how much food to buy, knowing how long between visits from family or friends | Much better | A bit better | Not much change | A bit worse | Much worse |
| 26. Using his/her intelligence to understand what's going on and to reason things through                                                  | Much better | A bit better | Not much change | A bit worse | Much worse |

APPENDIX 2

Percentile rank equivalents of IQCODE scores for various age groups

| Age group |       |       |       |     | Age group |       |       |       |     |
|-----------|-------|-------|-------|-----|-----------|-------|-------|-------|-----|
| Score     | 70-74 | 75-79 | 80-84 | 85+ | Score     | 70-74 | 75-79 | 80-84 | 85+ |
| 3-00      | 30    | 23    | 16    | 9   | 4-05      | 96    | 94    | 82    | 75  |
| 3-05      | 37    | 30    | 20    | 14  | 4-10      | 96    | 95    | 84    | 76  |
| 3-10      | 43    | 35    | 28    | 20  | 4-15      | 97    | 96    | 86    | 77  |
| 3-15      | 55    | 48    | 37    | 25  | 4-20      | 97    | 96    | 88    | 78  |
| 3-20      | 62    | 55    | 44    | 30  | 4-25      | 98    | 96    | 89    | 78  |
| 3-25      | 64    | 63    | 47    | 33  | 4-30      | 98    | 96    | 89    | 78  |
| 3-30      | 68    | 67    | 49    | 35  | 4-35      | 98    | 96    | 89    | 81  |
| 3-35      | 73    | 73    | 56    | 40  | 4-40      | 98    | 96    | 89    | 81  |
| 3-40      | 74    | 76    | 59    | 44  | 4-45      | 99    | 96    | 89    | 84  |
| 3-45      | 77    | 77    | 60    | 50  | 4-50      | 100   | 97    | 90    | 84  |
| 3-50      | 82    | 82    | 62    | 58  | 4-55      | 100   | 97    | 90    | 85  |
| 3-55      | 84    | 85    | 64    | 59  | 4-60      | 100   | 98    | 90    | 87  |
| 3-60      | 85    | 85    | 66    | 61  | 4-65      | 100   | 99    | 90    | 90  |
| 3-65      | 86    | 88    | 73    | 62  | 4-70      | 100   | 99    | 92    | 91  |
| 3-70      | 87    | 88    | 73    | 63  | 4-75      | 100   | 99    | 93    | 91  |
| 3-75      | 89    | 89    | 73    | 67  | 4-80      | 100   | 99    | 94    | 91  |
| 3-80      | 91    | 90    | 73    | 68  | 4-85      | 100   | 99    | 96    | 95  |
| 3-85      | 92    | 91    | 78    | 68  | 4-90      | 100   | 99    | 97    | 96  |
| 3-90      | 93    | 93    | 78    | 72  | 4-95      | 100   | 99    | 98    | 98  |
| 3-95      | 94    | 93    | 80    | 72  | 5-00      | 100   | 100   | 100   | 100 |
| 4-00      | 95    | 93    | 81    | 74  |           |       |       |       |     |